Eccentric Pieces

Why DICTO's sharp-edged pieces keep growing while the round stars freeze: the valley law



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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. In Star Cores of DICTO's Solids (#79) a star was built on a star: every triangle of the surface gets a spike, mirrored or golden, and DICTO's valley rule shortens each spike just enough to clear the pieces already built and its neighbours. The symmetric shapes freeze after two steps, while seven of DICTO's pieces keep growing: Lahdeli, Chucky, the Triskelion Tetrahedron, the Dodeca-13 Wedge, Samson, Fang and Kagemusha. We explain this by a theorem on a single edge. Cut the solid into pyramids from its centre; at an edge e the two pyramids meet the faces at angles α and α′, and the solid's dihedral angle there is $\delta = \alpha + \alpha'$. While the spikes at e are full, the spike of step k meets e at an angle γ_k with $\tan \gamma_k = (t - 1)^k \tan \alpha$, where t is the spike factor; the two spikes of step k can both be full only if δ and all the γ's on both sides fit in 360°; and once the two valley faces at e lie in one plane, nothing grows on them again. For mirrored spikes (t = 2) every γ_k equals α, so step k needs $(k + 1)\delta \le 360°$: an edge keeps its valley open through step two only if $\delta \le 120°$, through step three only if $\delta \le 90°$. For golden spikes the angles shrink by 1/φ each step, and a valley with equal sides stays open for ever if $\delta \le 112.20°$. Across the 50 shapes of the valley runs this one number, the sharpest dihedral angle, sorts the growth: each of the 14 convex shapes with an edge of 120° or less still adds 16 per cent or more of volume at step three, and each of the 30 with none adds at most 4.2 per cent (26 of them less than 0.1 per cent). Six of the seven pieces have edges of 90° or less. Kagemusha is non-convex: by a corollary of the law, the faces at its five reflex edges never take a spike, and it grows on its convex edges and ridges instead. DICTO's notch pieces fill the notches: with Maedate the star grows most, and with Kuwagata the star of mirrored spikes stops growing at step three. Whether a star keeps growing thus depends on its sharp edges and notches and on the rule used there. The formula is checked against the valley runs to 4.4×10^{-15} rad, and none of 676 closed valleys grows again.

1. Introduction

Put a spike on every face of a solid, then a spike on every face of the spikes, and again. Where two spikes lean towards each other, the valley between them fills up after a few steps. Whether it fills depends on one angle: how sharp the edge under the valley is.

The Star Cores papers follow DICTO's spiked solids. In The Truncated Icosahedron Star (#78) DICTO turned a solid's own face pyramids outwards. Star Cores of DICTO's Solids (#79, paper A) gave every new shape of DICTO Solids that treatment, and then built a second and third star on the first. With the tips straight over the triangles' centres, neighbouring spikes crossed in the valleys at the second step. DICTO proposed shortening the spikes in the valleys and keeping them full on the ridges: his valley rule. With it every run ended without a clash,

and the runs split in two. The symmetric shapes froze: at the third step they added almost nothing. Some of DICTO's pieces, the ones he built as parts of a fill rather than as whole symmetric solids, kept growing. Paper A named them and left them to this paper: Lahdeli, Chucky, the Triskelion Tetrahedron, the Dodeca-13 Wedge, Samson, Fang and Kagemusha. Core Cycles (#81, paper B) and The Gains of DICTO's Growth Rule (#80, paper C) dealt with the cores and the hulls of the chain. This paper deals with the eccentric pieces.

We prove a law for one edge of the solid, the valley law, and show that it sorts the growth of all 50 shapes of the valley runs by a single number, the solid's sharpest dihedral angle. The pieces keep growing because they have sharp edges. Symmetry plays no part in it: Samson and the Triskelion Tetrahedron have the full tetrahedral symmetry and keep growing, because their sharpest edges are right angles.

2. The seven pieces

Each piece comes from one of DICTO's papers and is in DICTO Solids v6 with its D-number. **Lahdeli** (D77) and Sonsam are the two pieces of the twinned octet at its equal-volume point (#56). **Chucky** (D80) and **Fang** (D81) are the pieces of the holes of the stacking ABAC, and the **Triskelion Tetrahedron** (D82) the cubic tetrahedral piece of Samson's family (#57, The Rule of Halves). **Samson** (D71) is the octet's tetrahedral piece at the Delian bond (#51). The **Dodeca-13 Wedge** (D6) is one of the two gap pieces of the DICTO Dodeca-13 (#13). **Kagemusha** (D86, earlier Shadow-17) is two of Johnson's gyroelongated pentagonal pyramids (J11) glued on their pentagons (#62); it is the only non-convex one.

D	Piece	Symmetry	V, E, F	Dihedral angles δ	Edges δ ≤ 120°	δ ≤ 90°	Open for ever at φ
D77	Lahdeli	D_{3d}	12, 18, 8	70.53°, 109.47°	18	6	12
D80	Chucky	C_{3v}	19, 30, 13	90°, 109.47°, 120°, 144.74°	24	6	6
D82	Triskelion Tetrahedron	T_d	16, 24, 10	90°, 125.26°	12	12	12
D6	Dodeca-13 Wedge	C_{2v}	8, 12, 6	10.31°, 85.83°, 91.60°, 107.82°, 108.22°	12	5	4
D71	Samson	T_d	16, 24, 10	90°, 125.26°	12	12	12
D81	Fang	C_{3v}	7, 12, 7	70.53°, 109.47°	12	3	6
D86	Kagemusha	D_{5h}	17, 45, 30	138.19°, 201.63° (5 reflex edges)	0	0	0

Table 1. The seven pieces. Dihedral angles to 0.01°, measured inside the solid. The last three columns count edges: valleys open through step two and step three at t = 2 (Theorem, (d)), and valleys that never close at t = φ.

3. Building on the star

Let S be a solid with its centre O (the mean of its corners) strictly inside every face plane, cut from O into one pyramid per face. A **step** puts a spike on every face of the current surface. For a face with the piece Q beneath it (at the first step the centre pyramid, later the spike it belongs to), let A be the point of Q farthest below the face's plane (the mean of such corners if there are several) and A_0 the foot of the perpendicular from A to the plane. The spike of height s is the pyramid over the face with tip

$$A + s\,(A_0 - A),$$

so s = 1 is flat, and the **full** spike has s = t, the spike factor: t = 2 mirrors Q's far point in the face, t = φ is the golden spike. At the first step A = O and this is #78's turn. **DICTO's valley rule** (#79) gives each spike the largest s in [1, t] for which it overlaps no piece already built; then any two new spikes that overlap are both cut back to the largest common s at which they do not. Overlaps are tested by separating axes in float64 (study 2026-10-30-star-cores, star_valley.py). We follow the rule exactly as it was run.

Every face of the solid contributes one triangle per edge to the surface after step one: the side of its spike over that edge. So every edge e of S carries two **valley faces**, one from each side, and after each later step the spike on a valley face has again exactly one face through e. The valley at e is the open wedge between the two valley faces.

4. Main theorem: the valley law

For an edge e between faces f and f′, let α be the angle at e between f and the plane through e and O, and α′ the same on the side of f′. The plane through e and O splits the solid's dihedral angle at e, so δ = α + α′. In the face plane of f, let d be the distance of the plane from O and ρ the signed distance from the foot of O to the line of e (positive when the foot lies on the face's side of e); then α = atan2(d, ρ), which exceeds 90° when the foot of O lies beyond e.

> **Lemma 1 (the angle of a spike at an edge).** Let a face F with piece Q beneath it contain the segment e, and let β_Q be Q's dihedral angle at e, between F and the plane through e and A. Then the spike of height s on F has the dihedral angle γ at e with
>
> $$\tan\gamma = (s - 1)\tan\beta_Q,$$
>
> taken in (0°, 180°), and it lies inside the wedge of angle γ on the far side of F from Q.

Proof. Take coordinates with e on the x-axis and F in the plane z = 0, Q below. Write $A_0 = (x, y, 0)$ and $A = (x, y, -h)$ with h > 0; then $\tan\beta_Q = h/y$ (with atan2 when y ≤ 0). The tip is $A + s(A_0 - A) = (x, y, (s - 1)h)$, so the spike's face through e makes the angle atan2((s − 1)h, y) with F. The spike is the hull of F and a point above F, so it lies in the wedge between F and that face. ∎

> **Theorem (the valley law).** Let S be convex with O strictly inside, e an edge with angles α, α′ and dihedral angle δ = α + α′, and t > 1 the spike factor. Suppose that the spikes on the two valley faces at e were full at steps 1, …, k − 1.
>
> (a) The full spike of step k on f's side meets e at the angle γ_k with $\tan\gamma_k = (t - 1)^k \tan\alpha$, and likewise γ'_k on the side of f′.
>
> (b) Both spikes of step k at e can be full only if
>
> $$\beta_k = 360° - \delta - (\gamma_1 + \ldots + \gamma_k) - (\gamma'_1 + \ldots + \gamma'_k) \geq 0.$$
>
> If $\beta_k < 0$, the valley rule shortens at least one of them.
>
> (c) If after some step the two valley faces at e lie in one plane, every later spike on either of them has height zero: the valley is closed for good.

(d) For $t = 2$, $\gamma_k = \alpha$ and $\gamma'_k = \alpha'$ for every k, so $\beta_k = 360° - (k + 1)\delta$: the two spikes of step k at e can be full only if $\delta \le 360°/(k + 1)$, that is $\delta \le 120°$ at step two and $\delta \le 90°$ at step three, whatever the split of δ into α and α'. For $1 < t < 2$ the angles shrink: at $t = \varphi$ and for $\alpha = \alpha'$, the valley takes two full spikes at step two if and only if $\tan^2(\delta/2) \le 4 + 2\sqrt{5}$ ($\delta \le 142.08°$), at step three if and only if $\delta \le 128.17°$, and at every step if and only if $\delta \le 112.20°$.

Proof. (a) We find the piece beneath each valley face and apply Lemma 1. At step 1 the face is f itself, Q is the centre pyramid and $A = O$, so $\beta_Q = \alpha$ and $\gamma_1 = \operatorname{atan2}((t - 1)d, \rho)$. At step 2 the valley face is the side T_1 of f's spike Q_1 over e. The corners of Q_1 are those of f and the tip, which lies on T_1; so the far point A lies in the plane of f, the plane through e and A is the plane of f, and β_Q is Q_1's angle at e, which is γ_1. At step $k \ge 3$ the valley face T_{k-1} is the face through e of the tetrahedron Q_{k-1} with corners the two ends of e and the tips of steps $k - 2$ and $k - 1$; its only corner off T_{k-1} is the tip of step $k - 2$, so the plane through e and A is T_{k-2} and $\beta_Q = \gamma_{k-1}$. By Lemma 1 with $s = t$, $\tan \gamma_k = (t - 1) \tan \gamma_{k-1}$, and by induction $\tan \gamma_k = (t - 1)^k \tan \alpha$.

(b) Near an inner point of e each piece through e coincides with its dihedral wedge at e. Going round e from the solid, the side of f carries the wedges $\gamma_1, \dots, \gamma_{k-1}$ of the earlier full spikes, one after the other (Lemma 1: each spike lies beyond the face it stands on), and the side of f' the same in the other sense. The two full spikes of step k add γ_k and γ'_k. If the total exceeds $360°$, the last two wedges overlap near every inner point of e, so the two full spikes overlap, and the valley rule cuts at least one of them back.

(c) If the two valley faces lie in one plane Π, they are two triangles in Π on the same side of the line e, each with e as an edge, so they overlap in a region of Π next to e. A spike of positive height on one of them reaches beyond Π into the piece standing on the other, near that region, and overlaps it; the valley rule gives it height zero. The face is then kept as it is, and the same holds at every later step.

(d) For $t = 2$, $\tan \gamma_k = \tan \alpha$ with both angles in $(0°, 180°)$, so $\gamma_k = \alpha$, and $\beta_k = 360° - \delta - k\alpha - k\alpha' = 360° - (k + 1)\delta$. For $t = \varphi$ and $\alpha = \alpha' = \delta/2$, write $x = \tan \alpha$ and $u = t - 1$. At step two the condition is $\gamma_1 + \gamma_2 \le 180° - \alpha$. With $\tan \gamma_1 = ux$ and $\tan \gamma_2 = u^2x$, equality holds when $\tan(\gamma_1 + \gamma_2) = -x$, that is $(u + u^2)x/(1 - u^3x^2) = -x$, or $x^2 = (1 + u + u^2)/u^3$. For $u = 1$ this gives $x^2 = 3$, $\alpha = 60°$, $\delta = 120°$, as above; for $u = 1/\varphi$ it gives $x^2 = \varphi^3(1 + 1/\varphi + 1/\varphi^2) = 2\varphi^3 = 4 + 2\sqrt{5}$, $\delta = 142.0786°$. The left side grows with δ, so the valley is open below this value and closed above it. The bounds $128.1727°$ ($k = 3$) and $112.2017°$ (all k; the series of the γ_k converges because $\tan \gamma_k \le \varphi^{-k} |\tan \alpha|$) are the roots of $\beta_k = 0$, found by bisection to 10^{-10}. ∎

The rule's common cut-back of (b) is what produces the situation of (c): when two valley spikes are cut back only against each other, the largest common height at which they do not overlap is the one at which $\beta = 0$, and their faces through e then lie in one plane.

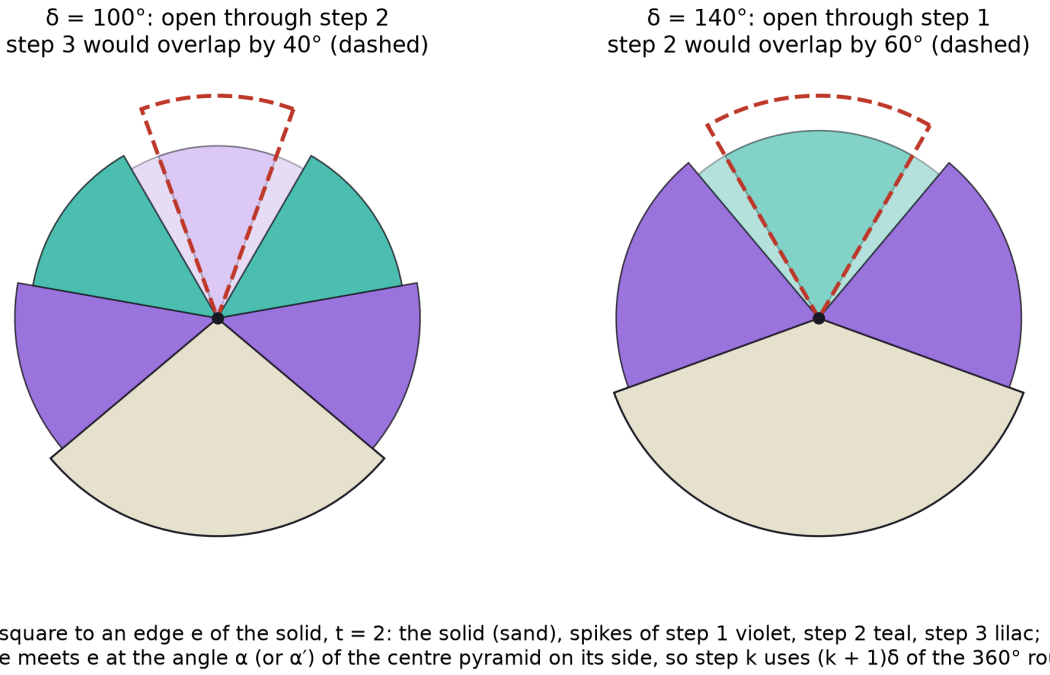


Figure 1. The valley law at t = 2, seen along an edge e of the solid. Each spike meets e at the angle α (or α′) of the centre pyramid on its side, so after k steps the solid and its spikes take (k + 1)δ of the 360° round e. Left: δ = 100°, open through step two; the spikes of step three would overlap by 40° (dashed). Right: δ = 140°, as on the symmetric shapes; the valley closes at step two.

5. The law against the runs

The formula. We rebuilt the valley rule exactly as in #79 for the seven pieces, five shapes for contrast (the Dodeca-13 Needle, the OCD Stella, the FPD, Tweedledum, the GTI) and the three skewed solids D1–D3, whose centre lies beyond some face edges (α > 90°), at t = 2 and t = φ, steps one to three, and measured at every edge of the solid the opening between its two valley faces. Wherever both valley spikes had been full, the measured opening agreed with β_k of the theorem to 4.4×10^{-15} rad. No edge with $\beta_k < 0$ had both spikes full. 676 valleys closed (386 at t = 2, 290 at t = φ), and none grew again.

The count. For all 50 shapes of the valley runs of #79 at both spike factors, every triangle of the surface after step one is a valley face of exactly one edge, so by (b) the number of full spikes at step two is at most twice the number of open edges plus the number of closed ones. This holds in every one of the 100 runs.

The split. Table 2 sorts the 44 convex shapes of the runs by their sharpest dihedral angle δ_{min}. The line at 120° is the line of (d) for step two at t = 2, and it separates the growth exactly.

Sharpest edge	Shapes	Full spikes at step 2 (t = 2)	Volume added at step 3 (t = 2)	Valleys open for ever at t = φ
$\delta_{min} \le 120°$	14	in every shape	16.5 to 57.5 per cent	in every shape (3 to 12 edges)
$\delta_{min} > 120°$	30	none	at most 4.2 per cent; less than 0.1 per cent in 26	none

Table 2. The 44 convex shapes of the valley runs of #79 (all with at most 200 faces). Sharp: the Dodeca-13 Wedge (10.31°), Ganesha (58.52°), the Dodeca-13 Needle (60.24°), the three skewed solids D1–D3 (63.43° to 79.19°), Sonsam, Lahdeli and Fang (70.53°), Samson, Ian

Dury, Blockhead, Chucky and the Triskelion Tetrahedron (90°). Round: the other 30, from Jubjub (135.52°) to Jabberwock (163.40°); the four that add more than 0.1 per cent are Walrus (4.2), Hatta (3.9), Haigha (2.3) and PHE (0.2). Data: valley-law-predict.json, star-valley.json.

The theorem proves the half of this that concerns the valleys: on a round shape every valley at an edge of the solid closes at step two at t = 2 and is never built on again. That the other faces of the round stars, the ridges and the faces round high corners, also stop growing at step three is observed in the runs, not proved here. On a sharp shape the theorem shows only that the valleys at the sharp edges stay open; that the rule then uses them is again what the runs show.

6. The pieces one by one

Piece	Full spikes, steps 1–3 (t = 2)	Steps 2–3, t = 2	Step 3, t = 2	Steps 2–3, t = φ
Lahdeli	8 / 36 / 12	×3.08	×1.335	×1.22
Chucky	13 / 42 / 24	×2.69	×1.436	×1.59
Triskelion Tetrahedron	10 / 24 / 0	×2.68	×1.386	×1.15
Dodeca-13 Wedge	6 / 14 / 12	×2.54	×1.470	×1.73
Samson	10 / 24 / 24	×2.47	×1.260	×1.19
Fang	7 / 12 / 0	×2.37	×1.372	×1.56
Kagemusha	20 / 0 / 5	×1.47	×1.123	×1.79
FPD (contrast)	60 / 0 / 0	×1.16	×1.000	×1.28

Table 3. The valley runs of #79 for the seven pieces and the FPD. Full spikes out of 8, 36, 108 for Lahdeli, and so on (the first count is the number of faces). Volume factors over the step before.

Lahdeli grows most at t = 2. All its edges are 70.53° or 109.47°, both under 120°, so every valley takes full spikes at step two: all 36 spikes of step two are full and the volume more than doubles (×2.31). At step three only its six edges of 70.53° stay open by the law ($4 \times 70.53° = 282° \leq 360° < 4 \times 109.47°$), and exactly 12 spikes are full, two per open edge.

Samson and the **Triskelion Tetrahedron** have the same dihedral angles, 90° at 12 edges and 125.26° at 12. At step two the law closes the valleys of the 125.26° edges and leaves the 90° ones; both pieces have exactly 24 full spikes, two per right-angled edge. At step three the right angles reach the bound of (d) exactly, $4 \times 90° = 360°$: the full spikes would just touch. Samson keeps all 24; in the Triskelion Tetrahedron other spikes reach them first, and none of step three is full, yet the step still adds 39 per cent.

Chucky has 6 edges of 90°, 3 of 109.47°, 15 of exactly 120° and 6 of 144.74°. At step two each of its 15 valleys of 120° takes both full spikes, which meet face to face ($\beta_2 = 0$), so by (c) they are closed for good; the six edges of 144.74° get no full spike; 42 of the 60 spikes are full. At step three none is full, but the step adds 44 per cent.

The **Dodeca-13 Wedge** has the sharpest edge of all the shapes, 10.31°, the thin edge of the wedge: by (d) its valley could stay open for 33 steps at t = 2. At t = φ it grows most after Kagemusha. **Fang** has three edges of 70.53° and nine of 109.47°: 12 full spikes at step two, none at step three, but still 37 per cent more volume, from spikes cut back by the rule.

Kagemusha is the exception. It is non-convex: the two J11 meet in five reflex edges of 201.63°, and its other 40 edges are 138.19°. The theorem is stated for convex solids, but its proof uses only the edge and the pieces next to it, and at a reflex edge (b) gives $\beta_1 = 360° - 2\delta < 0$ at t = 2: the valleys at the reflex edges cannot take two full spikes even at step one. In the runs ten faces stay flat at step one, at both spike factors. They stay flat at every step, and the notches never fill:

> **Corollary (empty notches).** Let e be an edge of a solid S with O inside, whose angles α and α′ both exceed 90° (the foot of O lies beyond e in both face planes), and suppose $\beta_1 < 0$. Then under the valley rule neither face at e ever takes a spike of positive height.

Proof. At step 1 the piece beneath each face is its centre pyramid and A = O, so by Lemma 1 the spike of height s on f meets e at $\gamma(s) = \operatorname{atan2}((s-1)d, \rho)$ with $\rho < 0$. As s falls from t towards 1, γ(s) *rises* from γ_1 towards 180°: shortening the spike turns it further over the notch. So for any heights s, s′ in (1, t], $\gamma(s) + \gamma'(s') \geq \gamma_1 + \gamma'_1 = 360° - \delta - \beta_1 > 360° - \delta$, and as in (b) the two spikes overlap near every inner point of e. The rule's common cut-back finds no positive height at which they are clear, and gives both height zero. Each face then keeps its centre pyramid beneath it and its flat partner beside it, so the same holds at every later step. ∎

At a convex edge the opposite holds: $\rho > 0$, γ(s) falls to 0 as $s \to 1$, and the cut-back always finds a positive height; that is where the growth from cut-back spikes comes from. At Kagemusha's reflex edges $\tan\alpha = \tan\alpha' = -(3+\sqrt{5})$, $\alpha = \alpha' = 100.81°$, and $\beta_1 = -43.25°$ at t = 2 and −55.97° at t = φ; in the runs the ten faces beside them stay flat at all three steps, and the notch stays open at exactly $360° - \delta = 158.38°$.

Kagemusha grows on its other 20 faces, the two caps (Table 4). Its 40 convex edges of 138.19° split in three ways: ten edges A with $\alpha = \alpha' = 69.09°$ ($\tan\alpha = \varphi^2$), ten B with 49.79° and 88.40°, and twenty C, each beside a notch face, with 53.64° and 84.55°. At t = 2 no valley can take two full spikes at step two by (d), and none does: the step adds 31 per cent, all from spikes cut back by the rule, as for Fang. At t = φ the law itself leaves the ten A valleys open at step two (138.19° < 142.08°, $\beta_2 = +15.25°$), and all 20 of their spikes are full. The C valleys are one-sided, since the notch face beside each never takes a spike, and their 20 spikes are full as well; the B valleys ($\beta_2 = -11.91°$) are cut back. At step three the A valleys close ($\beta_3 = -48.19°$), and most of the growth is on the ridges of the cap spikes: 60 per cent at t = 2, 93 per cent at t = φ. This is why Kagemusha grows most of all the pieces at t = φ (×1.79 over steps two and three).

t, step	Volume ×	Notch faces	A valleys	B valleys	C valleys	Ridges
2, step 2	1.3115	0	52.9 % (0 / 20 / 0)	15.3 % (0 / 20 / 0)	31.8 % (0 / 9 / 11)	—
2, step 3	1.1227	0	0	0	39.9 %	60.1 %
φ, step 2	1.5045	0	43.6 % (20 / 0 / 0)	23.1 % (0 / 20 / 0)	33.3 % (20 / 0 / 0)	—
φ, step 3	1.1889	0	0	0	6.8 %	93.2 %

Table 4. Where Kagemusha's volume goes: share of the volume gained at each step, with full / cut-back / flat spikes in brackets. At step one the 20 cap faces take full spikes and the 10 notch faces none (×1.7405 at t = 2, ×1.4577 at t = φ). The spike volumes add up to the volume gained to 7×10^{-15} (study 2026-11-01-star-cores-d, kagemusha_notches.py).

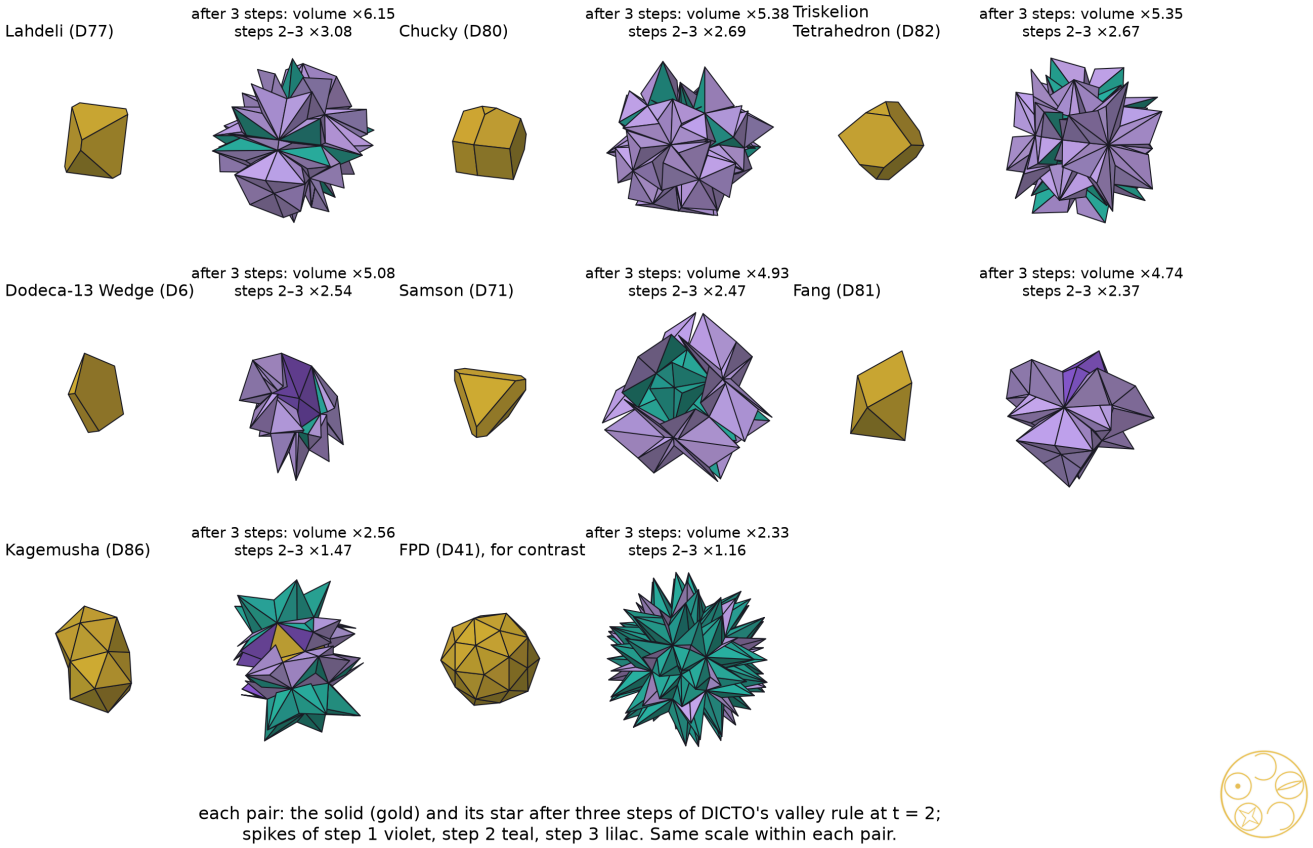


Figure 2. The seven eccentric pieces and the FPD (D41) for contrast: each solid (gold) and its star after three steps of DICTO's valley rule at t = 2, at the same scale within each pair. Spikes of step one violet, step two teal, step three lilac. Volumes as multiples of the solid's, and the growth over steps two and three.

DICTO's notch variants: Maedate and Kuwagata

Under the valley rule Kagemusha's notches stay empty. DICTO proposed two ways to fill them at step one and named the pieces. **Maedate** (前立, the front crest of a samurai helmet) joins the two crossing notch spikes into one piece: the hull of the two notch faces (a, b, c) and (a, b, c′) and their full tips T and T′, one per notch, an irregular octahedron (6 corners, 8 triangles). **Kuwagata** (鍬形, the V-shaped horns of the helmet) follows DICTO's sloping-base rule: each notch face keeps a pyramid, but its apex is lowered along the line from c to T until it meets the notch's mirror plane, the plane of the shared pentagon, so the two pyramids meet face to face; the pair is an irregular triangular dipyramid. In every run both pieces lie outside the solid and clear of every other piece. The 20 cap faces take full spikes at step one as before, and steps two and three follow the valley rule unchanged (Table 5, Figure 3).

Notch rule	t	Step 1	Step 2	Step 3	After three steps	Notch piece volume
valley rule (empty notches)	2	×1.7405	×1.3115	×1.1227	2.563	0
Maedate	2	×2.0354	×1.3216	×1.1368	3.058	0.2218
Kuwagata	2	×1.9404	×1.3102	×1.0000	2.542	2 × 0.0752
valley rule (empty notches)	φ	×1.4577	×1.5045	×1.1889	2.607	0
Maedate	φ	×1.6236	×1.4830	×1.1032	2.656	0.1248
Kuwagata	φ	×1.5869	×1.4521	×1.1784	2.715	2 × 0.0486

Table 5. Kagemusha's star with its notches left empty and with DICTO's two notch pieces: volume factors over the step before, the volume after three steps and the volume of the notch piece at each notch, all with Kagemusha's volume taken as 1. Every surface is closed with

Euler characteristic 2, and the volumes add up exactly (study 2026-11-01-star-cores-d, kagemusha_variants.py).

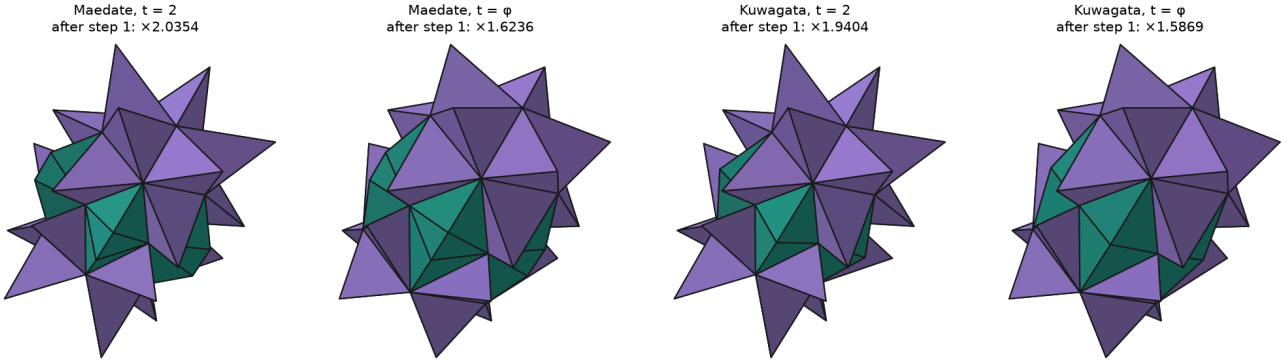


Figure 3. Kagemusha after step one with DICTO's notch pieces, Maedate and Kuwagata (teal), at t = 2 and t = φ, each at its own scale; the cap spikes are violet.

Both pieces fill the notches. Maedate makes Kagemusha grow most at t = 2 (×3.058 after three steps, against ×2.563 with empty notches), and its star keeps growing at both spike factors. With Kuwagata at t = 2 the star stops: step three gives ×1.000000000, as the ten spikes still cut back add no volume to nine figures. At t = φ Kuwagata gives the largest star of the three, ×2.715. An analogy, and no more than that: in crystal growth the re-entrant corner between the two halves of a twin is where new layers start most easily, and twinned germanium dendrites grow from such corners (D. R. Hamilton and R. G. Seidensticker, 1960). Maedate and Kuwagata turn Kagemusha's re-entrant edges, where the valley rule lets nothing grow, into places that grow at the first step. Whether the new forms close into a cycle, and with what period, has not been computed.

7. What is new and what is known

- **Known, credited:** DICTO's turn and its spikes (#78); the building on the star, DICTO's valley rule and the observation that the symmetric shapes freeze while the eccentric pieces keep growing (#79); the pieces and their dihedral angles (#13, #51, #56, #57, #62; DICTO Solids v6). That copies of a wedge fit round an edge only while their angles add up to at most 360° is elementary; the gap left by five regular tetrahedra round an edge is its best-known case (J. C. Lagarias, C. Zong, "Mysteries in packing regular tetrahedra", Notices AMS 59 (2012) 1540–1549). That gluing pyramids on faces can make new edges sharper than the old is used by D. Eppstein for iterated Kleetopes (arXiv:2009.00116).
- **New, as far as the sources checked show:** the valley law, that the k-th spike at an edge meets it at $\tan \gamma_k = (t-1)^k \tan \alpha$, with the closing condition $\beta_k \geq 0$ and the closed-for-good property; the mirrored bound $\delta \leq 360°/(k+1)$, independent of how δ splits; the golden bounds 142.08° ($\tan^2(\delta/2) = 4 + 2\sqrt{5}$), 128.17° and 112.20°; the sorting of all 44 convex shapes of the runs by their sharpest dihedral angle; the corollary that a notch with both feet of O beyond its edge and $\beta_1 < 0$ stays empty for ever, and the account of Kagemusha's growth; DICTO's notch pieces Maedate and Kuwagata, and the star of Kagemusha with Kuwagata at t = 2, which stops growing at step three.
- **Not decided here:** a proof that the ridges and corner faces of the round stars stop growing; a law for the growth from cut-back spikes at t = 2 (Fang, Kagemusha) and on the ridges at step three (Kagemusha); whether the sharp shapes grow without end under the valley rule.

Sources checked (1 November 2026). *Reference library* (ask-library on dicto-big-brother, top five hits each): "iterated stellation by adding pyramids on every face repeatedly, growth of a polyhedron" (Messer 1995, Kunnen–Schlicker 1998, Richter-Gebert 2026, Wikipedia "Goldberg polyhedron", Kabai 2010; pages Messer 1995 pp. 1–2, Torquato–Jiao arXiv:1206.3716 p. 21): stellation of face planes, not pyramids built on pyramids; "dihedral angle 120 degrees reflection of a tetrahedron across its face, packing tetrahedra around an edge" (Richter-Gebert 2026, Torquato–Jiao 2009, Sommerville 1922, Wikipedia "Rhombic dodecahedron", Lagarias–Zong 2012; pages Kepler 1619 p. 271, Hynd 2023 p. 28, Frettlöh 2011 p. 1, Koskinen 2011 p. 1, Bianchi–Doná–Speziale 2011 p. 6): the edge-angle count for tetrahedra, no spikes; "Kleetope iterated, augmenting faces with pyramids, self-intersection" (He–Guest 2025, Murphy–Weed 2024, O'Rourke 2010, Bernhard 2026, Kubala 2019; pages Krishnamurthy–Akleman–Subramanian 2020 p. 5, Grünbaum 2010 p. 13, Wikipedia "N-flake" p. 6, Akpanya et al. 2024 pp. 1, 27): nothing on the angle at an edge under repeated pyramids; "reflecting a polyhedron in its faces, mirror images around an edge, dihedral angle condition" (Richter-Gebert 2026, Bochis–Santos 2002, Montejano et al. 2019, Hynd 2023, Wikipedia "Rhombic dodecahedron"; pages Cutler–Schulte 2011 pp. 13, 15, 16, Torquato–Jiao 2009 p. 5, Kepler 1619 p. 271; note Fedorov 1885 ch. 13): nothing on stacked mirror images round an edge. *Web* (one search, "iterated Kleetope repeated pyramid augmentation of faces self-intersection dihedral angle condition"): D. Eppstein, "On polyhedral realization with isosceles triangles", arXiv:2009.00116 (Graphs and Combinatorics 2021), summary and abstract only; Wikipedia "Triakis tetrahedron" and "Small hexagonal hexecontahedron" (titles only). No source states the valley law.

8. Conclusion

A spike on a valley face leans on the spike beneath it, and the angle it makes with the solid's edge is the angle of the piece below, stretched by the spike factor. For mirrored spikes it is exactly the angle of the centre pyramid, step after step, so the spikes stack round each edge like copies of the same two wedges, and an edge of dihedral angle δ has room for $360°/\delta - 1$ steps. A round shape, all of whose edges are blunter than 120°, fills every valley at the second step, and a filled valley never opens again: the star freezes. DICTO's eccentric pieces are not round. Six of them have edges of 90° or less, and their valleys stay open while the round stars stop; the Dodeca-13 Wedge, with its edge of 10.31°, has room for 33 steps. So the pieces keep growing because their edges are sharp, not because their symmetry is low: Samson and the Triskelion Tetrahedron are as symmetric as the tetrahedron. Kagemusha, with no sharp edge but five reflex ones, grows on its caps: under the valley rule its notches stay empty for ever, and its growth is cut-back spikes, the open golden valleys of step two and the ridges of step three. DICTO's notch pieces change this. With Maedate, the conjoined crest, the notches fill and the star grows most at $t = 2$; with Kuwagata, his sloping bases, the star of mirrored spikes stops growing at step three, so the one non-convex piece also reaches a finished form. Whether a star keeps growing is therefore decided by the solid's sharp edges and notches together with the rule used there: the valley law says which valleys stay open, and the rule at the notches decides whether the star stops. Only Kagemusha was run with the notch pieces. The six convex eccentric pieces still grow under the valley rule, and whether sloping bases in their open valleys would stop them too is the natural next test, not made here.

Data and code. Study 2026-11-01-star-cores-d: scripts/valley_law.py (part 1: α, α', δ and β_k for every edge of the 50 shapes of the valley runs, with δ checked against the face normals to 10^{-9}; part 2: the valley rule of star_valley.py rebuilt and instrumented for the seven pieces and five contrast shapes at $t = 2$ and φ; data/valley-law-predict.json, valley-law-run.json, valley-law.log), valley_law_extra.py (the skewed solids D1–D3; valley-law-run-extra.json), figs_paper_d.py (Figures 1 and 2). It imports star_core.py, star_valley.py and figs_paper_a.py of study 2026-10-30-star-cores, whose star-valley.json gives the growth figures. DICTO Solids v6 (dataset concept DOI 10.5281/zenodo.23606967). Run on dicto-big-brother, float64.

Sources. DICTO, The DICTO Dodeca-13 (#13), DICTO DOI 10.5281/zenodo.23284281. DICTO, The Delian Bond (#51), DICTO DOI 10.5281/zenodo.23606828. DICTO, Sonsam and Lahdeli (#56), DICTO DOI

10.5281/zenodo.23635823. DICTO, The Rule of Halves (#57), DICTO DOI 10.5281/zenodo.23647898. DICTO, Shadows of the Fourth Dimension (#62), DICTO DOI 10.5281/zenodo.23687223. DICTO, The Truncated Icosahedron Star (#78, v2), DICTO DOI 10.5281/zenodo.23782577. DICTO, Star Cores of DICTO's Solids (#79), DICTO DOI 10.5281/zenodo.23822905. DICTO, The Gains of DICTO's Growth Rule (#80), DICTO DOI 10.5281/zenodo.23822907. DICTO, Core Cycles (#81), DICTO DOI 10.5281/zenodo.23846772. J. C. Lagarias, C. Zong, Notices AMS 59 (2012) 1540–1549. D. Eppstein, arXiv:2009.00116. D. R. Hamilton, R. G. Seidensticker, "Propagation mechanism of germanium dendrites", J. Appl. Phys. 31 (1960) 1165–1168, doi:10.1063/1.1735796.